



■ CS177 Bioinformatics: Software Development and Applications

Lecture 16: Genomics

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Learning objectives



- At the end of this lecture, you will be able to:
 - Define genomics and related basic concepts (e.g. functional genomics)
 - Name a few next-generation sequencing (NGS) technologies
 - Describe and explore some open data resources of genomics (e.g. cancer genomics projects)
 - Explain main ideas of genome-wide association studies (GWAS)
 - Explain the key idea of using convolutional neural network for modeling protein-DNA binding specificities
 - Describe what DeepBind can do and how it works





Overview of Genomics

PPT slides of this module are courtesy from Dr. Lichun Jiang, SIAIS, ShanghaiTech

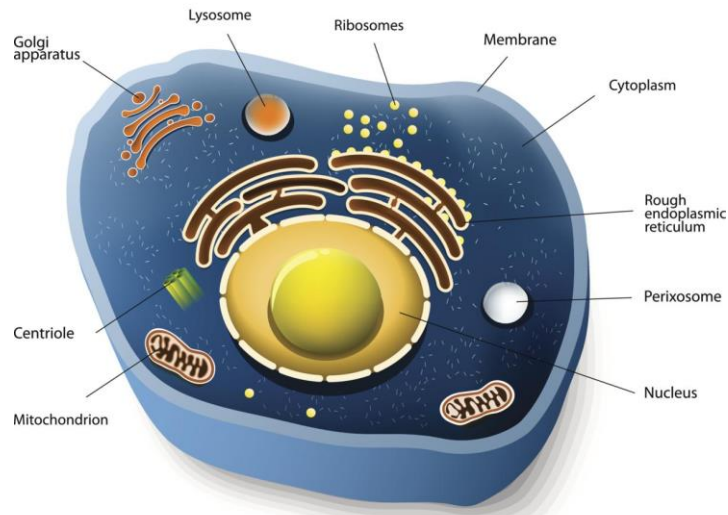




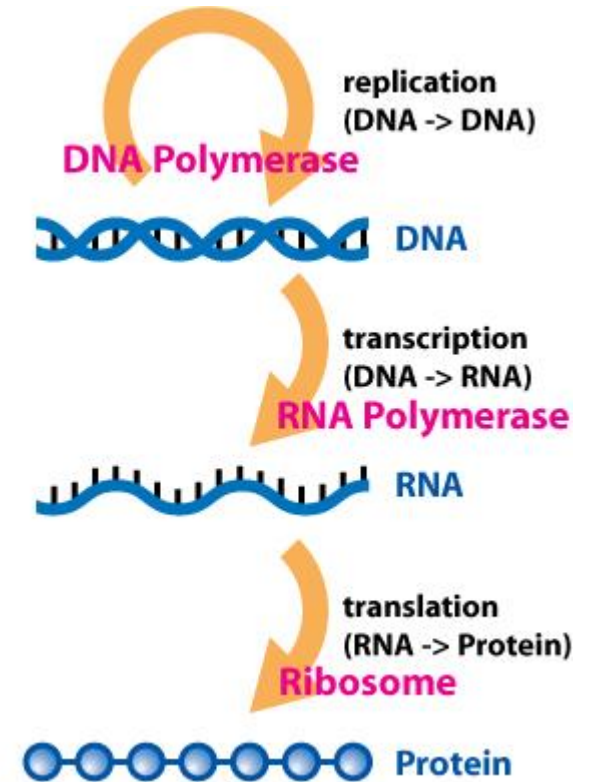
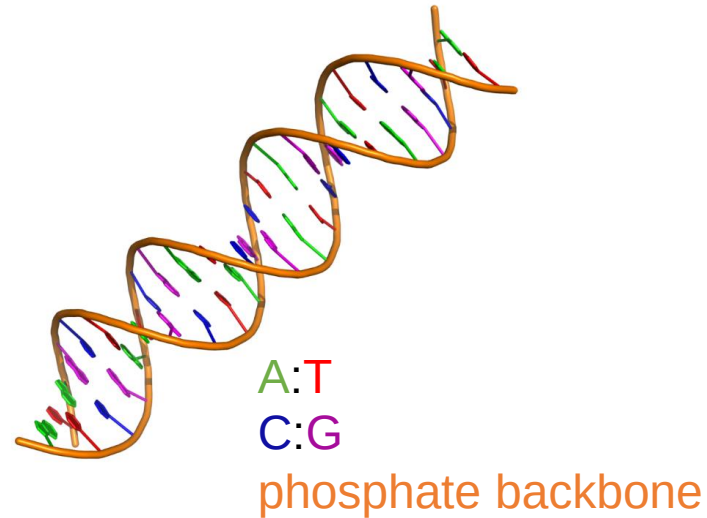
Information flow in cells: Central dogma of modern biology

DNA double helix

ANIMAL CELL



<https://media.buzzle.com/>



https://en.wikipedia.org/wiki/Central_dogma_of_molecular_biology

<https://commons.wikimedia.org/w/index.php?curid=80147407>



What are genome and genomics

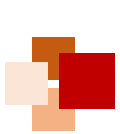


- An organism's complete set of DNA is called its **genome**. Virtually every single cell in our body contains a complete copy of the approximately 3 billion DNA base pairs, or letters, that make up the human genome.
- **Genomics** is an interdisciplinary field of biology focusing on the structure, function, evolution, mapping, and editing of genomes.

<https://ghr.nlm.nih.gov/primer/hgp/genome>

<https://en.wikipedia.org/wiki/Genomics>

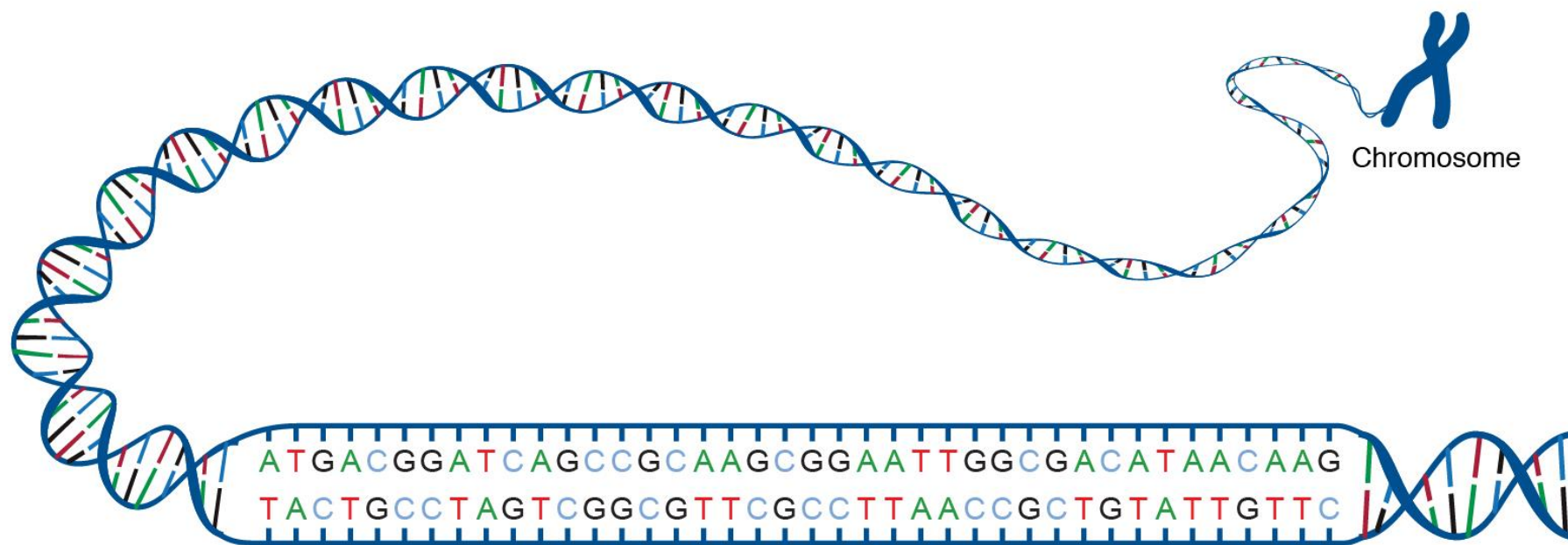




DNA sequencing: technology to read DNA



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<https://www.genome.gov/genetics-glossary/acgt>



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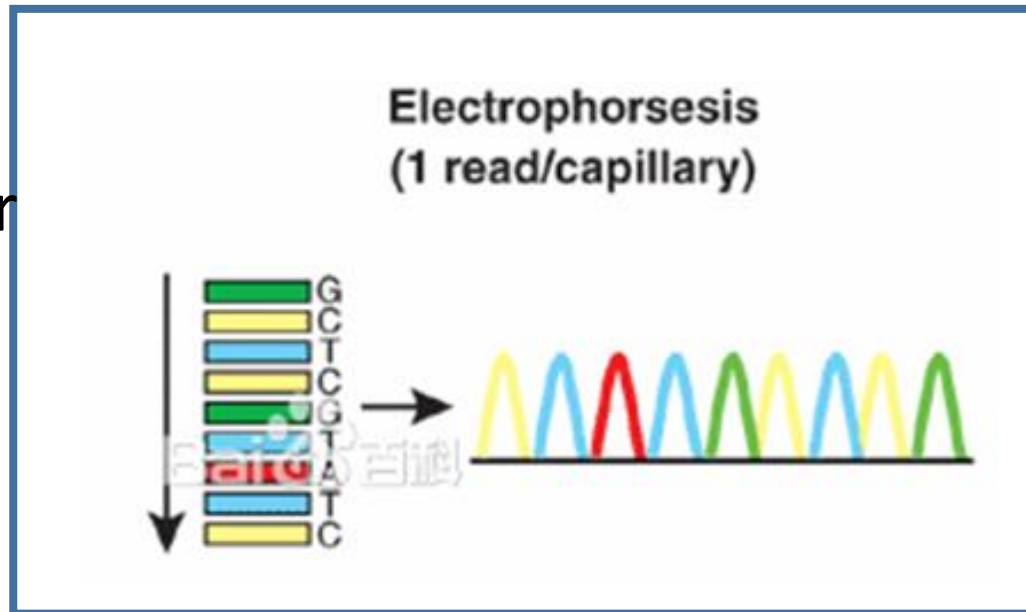


DNA sequencing technologies

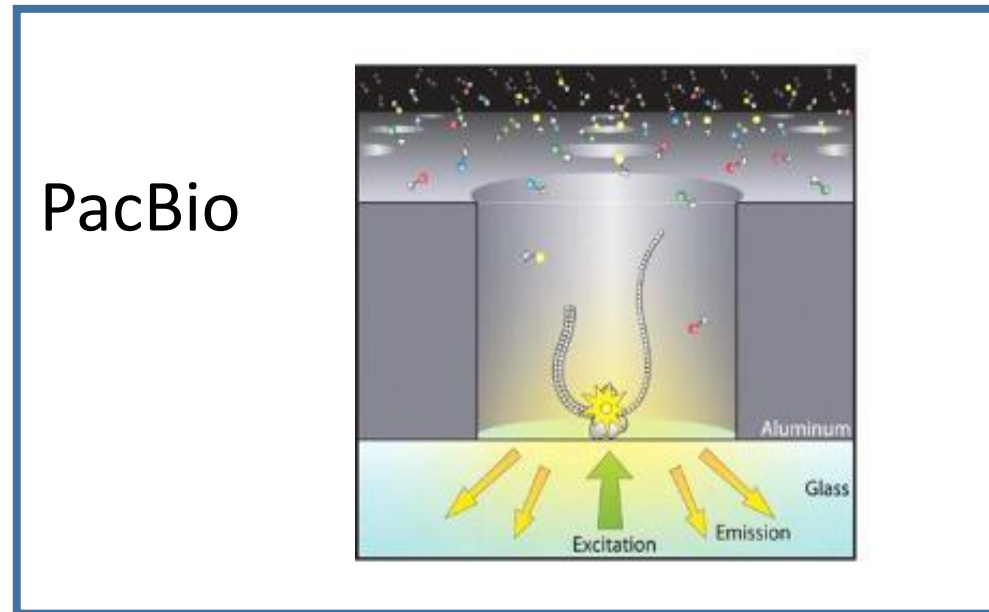


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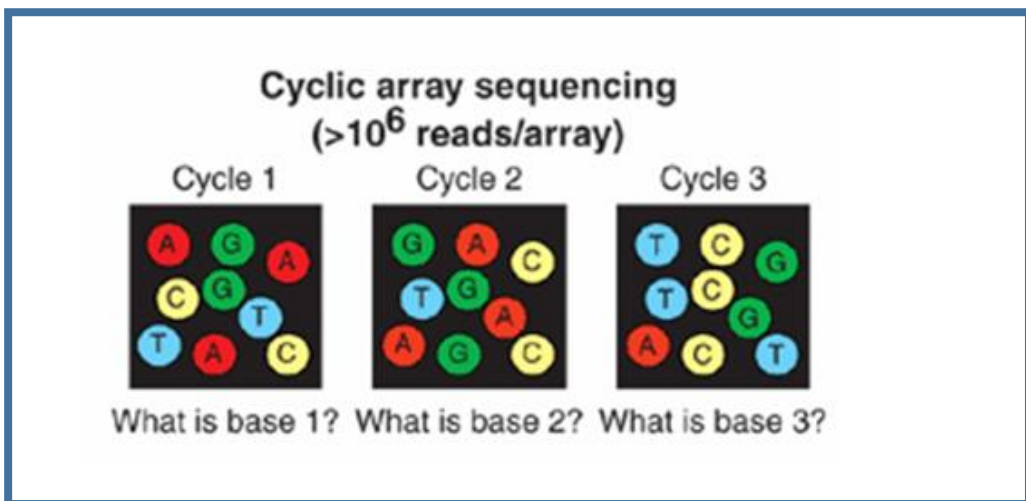
Sanger



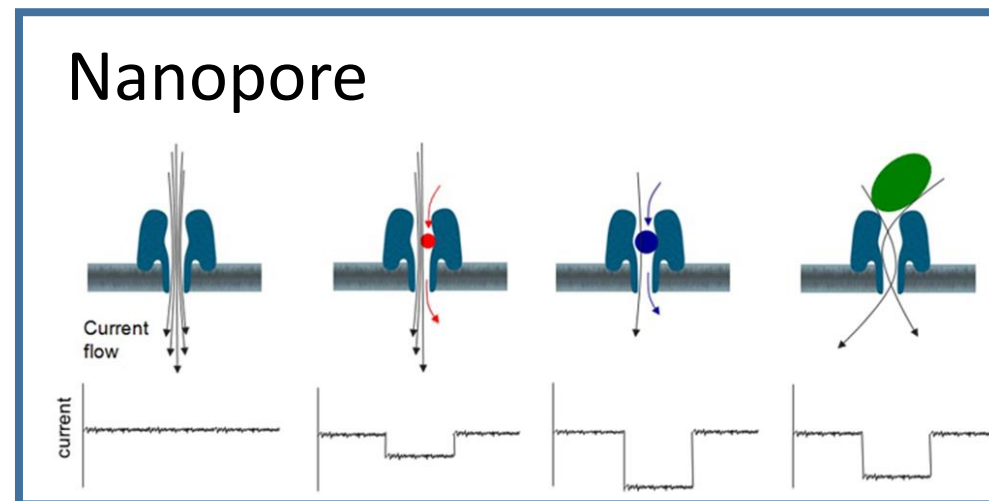
PacBio



NGS



Nanopore



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Next-generation sequencing (NGS)



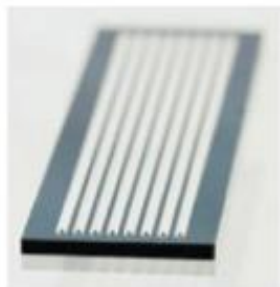
Technology	Read length (bp)	Reads per run	Time per run	Cost per megabase	Accuracy
Roche 454	700	1 million	1 day	\$10	99.9%
Illumina	50-250	<3 billion	1-10 days	~\$0.10	98%
SOLiD	50	~1.4 billion	7-14 days	\$0.13	99.9%
Ion Torrent	200	<5 million	2 hours	\$1	98%
Pacific Biosciences	2900	<75,000	<2 hours	\$2	99%
Sanger	400-900	N/A	<3 hours	\$2400	99.9%

Note: The above information is old, before 2018



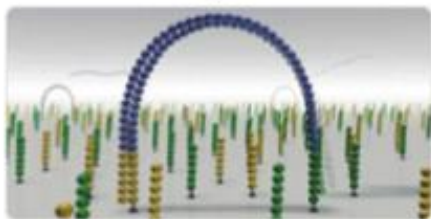
Randomly fragment genomic DNA

Library preparation



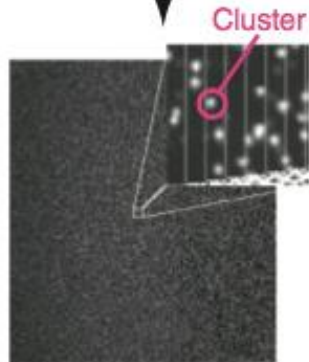
Samples immobilized on
surface of a flow cell (8 lanes)

Solid phase amplification



- Bridge amplification (inverted U) generates clusters on surface of flow cell
- ~Ten million single-molecule clusters per square centimeter

Sequencing by synthesis



- Each cycle: add polymerase, one labeled deoxynucleoside triphosphate (dNTP) at a time (four labeled dNTPs per cycle)
- Image fluorescent dyes
- Call nucleotide
- Enzymatic cleavage to remove

■ Sequencing by Illumina

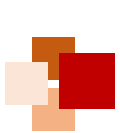
Disadvantage:

- Short read length (~150 bases)

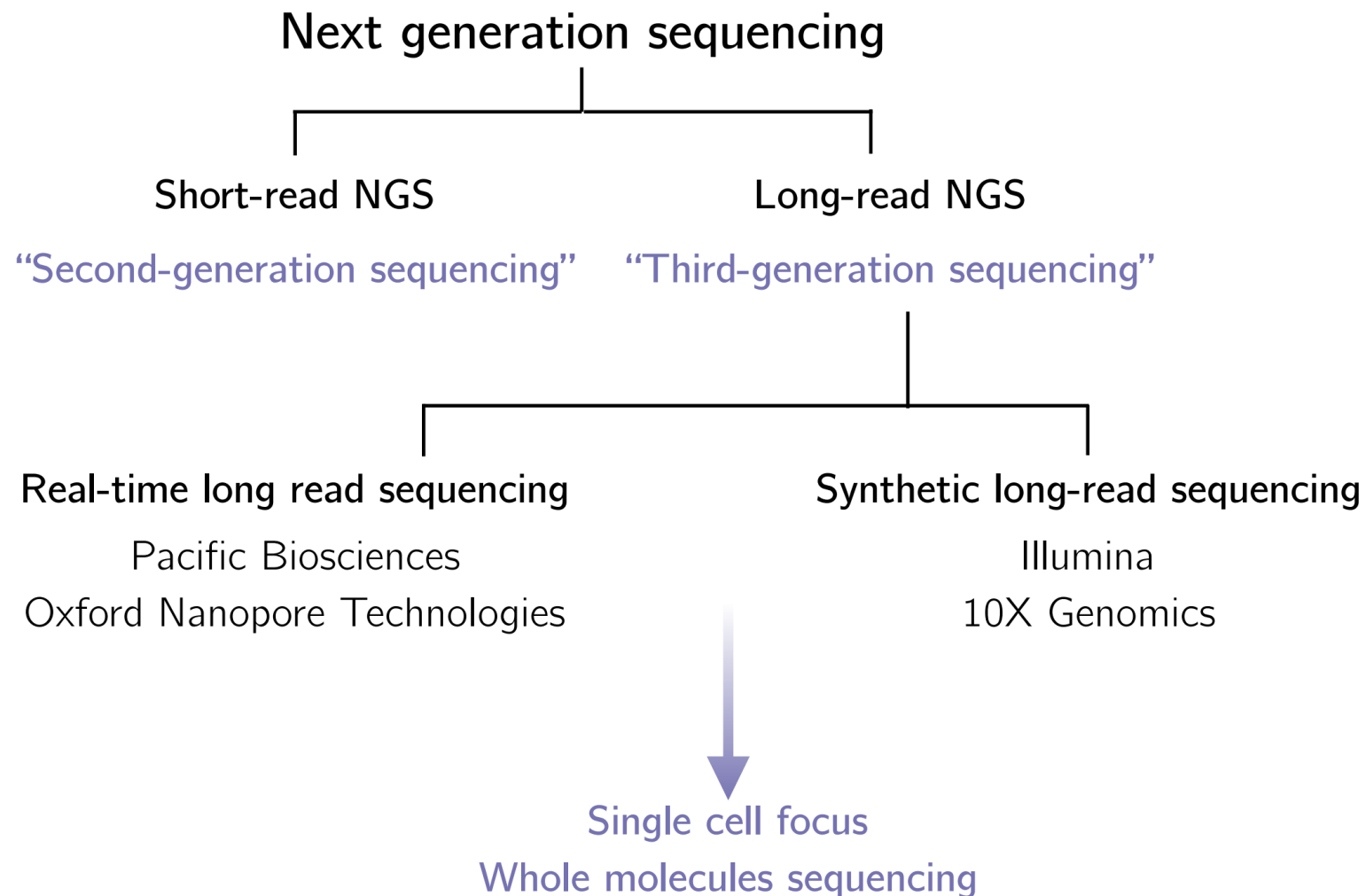
Advantages:

- Very fast
- Low cost per base
- Large throughput, up to 1 gigabase/experiment
- Short read length makes it appropriate for resequencing
- No need for gel electrophoresis
- High accuracy
- All four bases are present at each cycle, with sequential addition of dNTPs. This allows homopolymers to be accurately read

https://www.bilibili.com/video/BV1v34y147nj/?spm_id_from=333.788.recommend_more_video.3

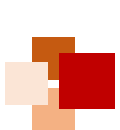


Third-generation sequencing (TGS)



Goodwin, S., McPherson, J. D., & McCombie, W. R. (2016). Coming of age: Ten years of next-generation sequencing technologies. *Nature Reviews Genetics*, 17(6), 333–351.



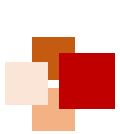


Third-generation sequencing (TGS)



- TGS is also known as long-read sequencing
 - In second-generation sequencing (SGS), DNA is fragmented into tiny bits, amplified and then sequence
 - TGS methods do not break or amplify DNA; instead, they sequence a single DNA molecule directly
- Strengths:
 - Offer capability of **single molecule real-time sequencing** of **longer** reads
 - Detect DNA modifications without any assay
 - Higher speed of sequencing than SGS, esp. in metagenomics
 - Better performance for transcript detection, mRNA abundance estimation, etc.
- Main challenge:
 - **High error rates**, due to instability of the molecular machinery involved
- Representative TGS technologies:
 - PacBio (Pacific Biosciences) SMRT (single molecule real-time) sequencing
 - Oxford nanopore sequencing





Third-generation sequencing (TGS)



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Aa Pacific Biosciences

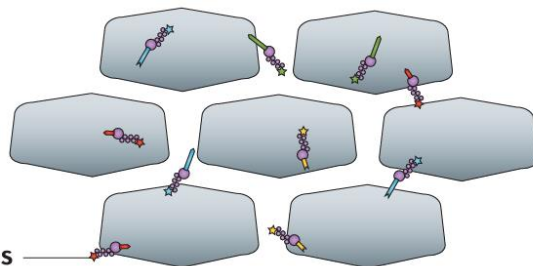
SMRTbell template

Two hairpin adapters allow continuous circular sequencing



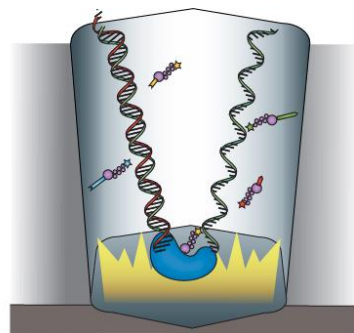
ZMW wells

Sites where sequencing takes place



Labelled nucleotides

All four dNTPs are labelled and available for incorporation



Modified polymerase

As a nucleotide is incorporated by the polymerase, a camera records the emitted light

PacBio output

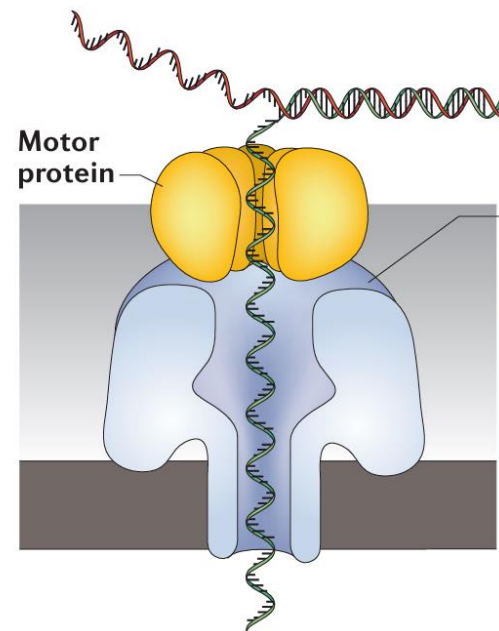
A camera records the changing colours from all ZMWs; each colour change corresponds to one base



Ab Oxford Nanopore Technologies

Leader-Hairpin template

The leader sequence interacts with the pore and a motor protein to direct DNA, a hairpin allows for bidirectional sequencing

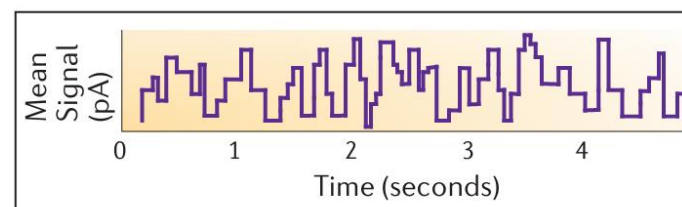


Alpha-hemolysin

A large biological pore capable of sensing DNA

Current

Passes through the pore and is modulated as DNA passes through

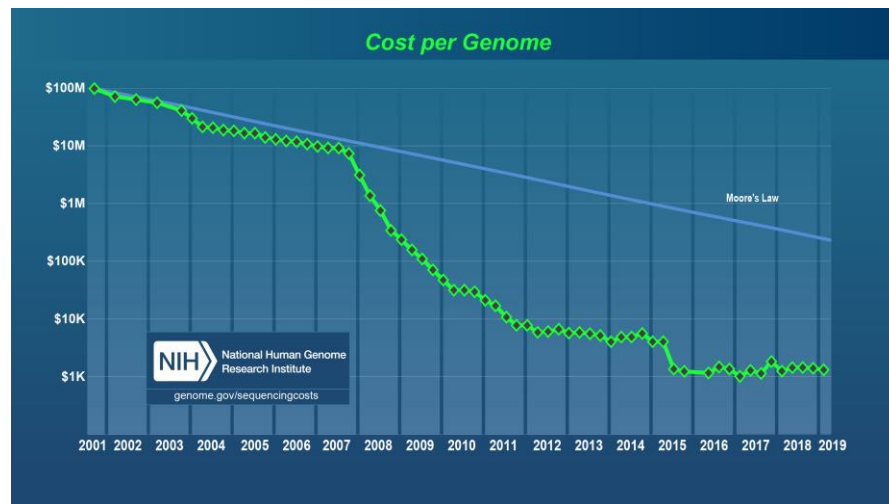


ONT output (squiggles)

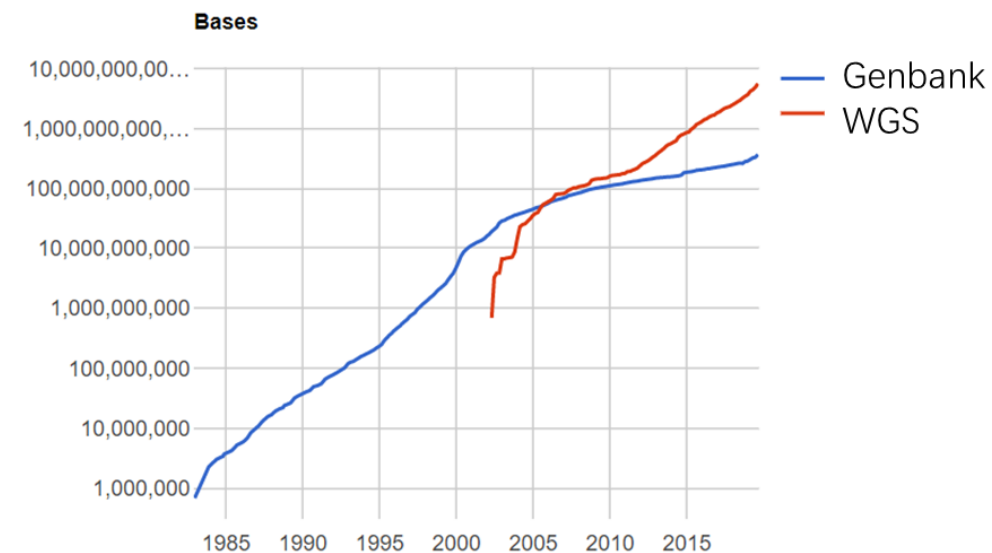
Each current shift as DNA translocates through the pore corresponds to a particular k-mer

Sara Goodwin et al. (2016) Coming of age: ten years of next-generation sequencing technologies, Nature Reviews Genetics.

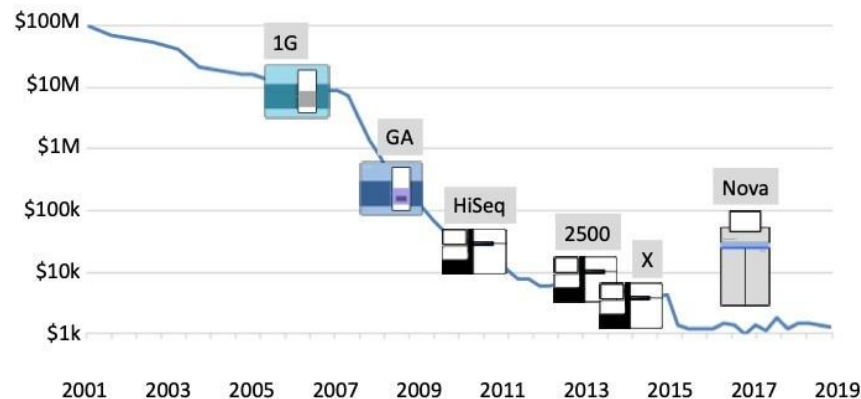
Cost of DNA sequencing dropped more than Moore's Law in the past decade



GenBank and WGS Statistics



Production cost per 30x Human genome over 18 years



<https://www.ncbi.nlm.nih.gov/genbank/statistics/>

<http://enseqlopedia.com/>



Data Resources of Genomics

PPT slides of this module are courtesy from Dr. Lichun Jiang, SIAIS, ShanghaiTech



It took 20 years from a human genome draft to a complete one



2001



2022

- International Human Genome Consortium. Initial sequencing and analysis of the human genome. Nature, 409: 860-921. 2001.
- Venter, J.C. Adams, M.D., Myers, E.W., Li, P.W., Mural, R.J., et al. The sequence of the human genome. Science, 291: 1304-1351. 2001.
- The complete sequence of a human genome. Science. 2022 March 376:44-53



FASTA

Sequencing raw reads

Mapping file

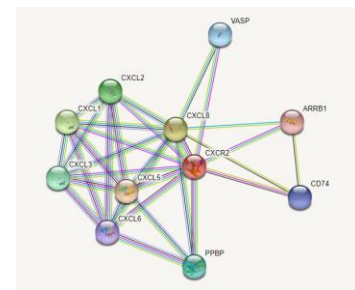
Bed format

Matrix format

```
[jianglc@SIAIS-mgt Quantification]$ head -n3 expression.txt
```

Gene_ID	Reads_T1	Reads_T2	Reads_T3	Reads_C1	Reads_C2	Reads_C3
ENSRNOG00000047964	0	2	20	108	201	189
ENSRNOG00000050370	11	13	20	900	1003	980

Network

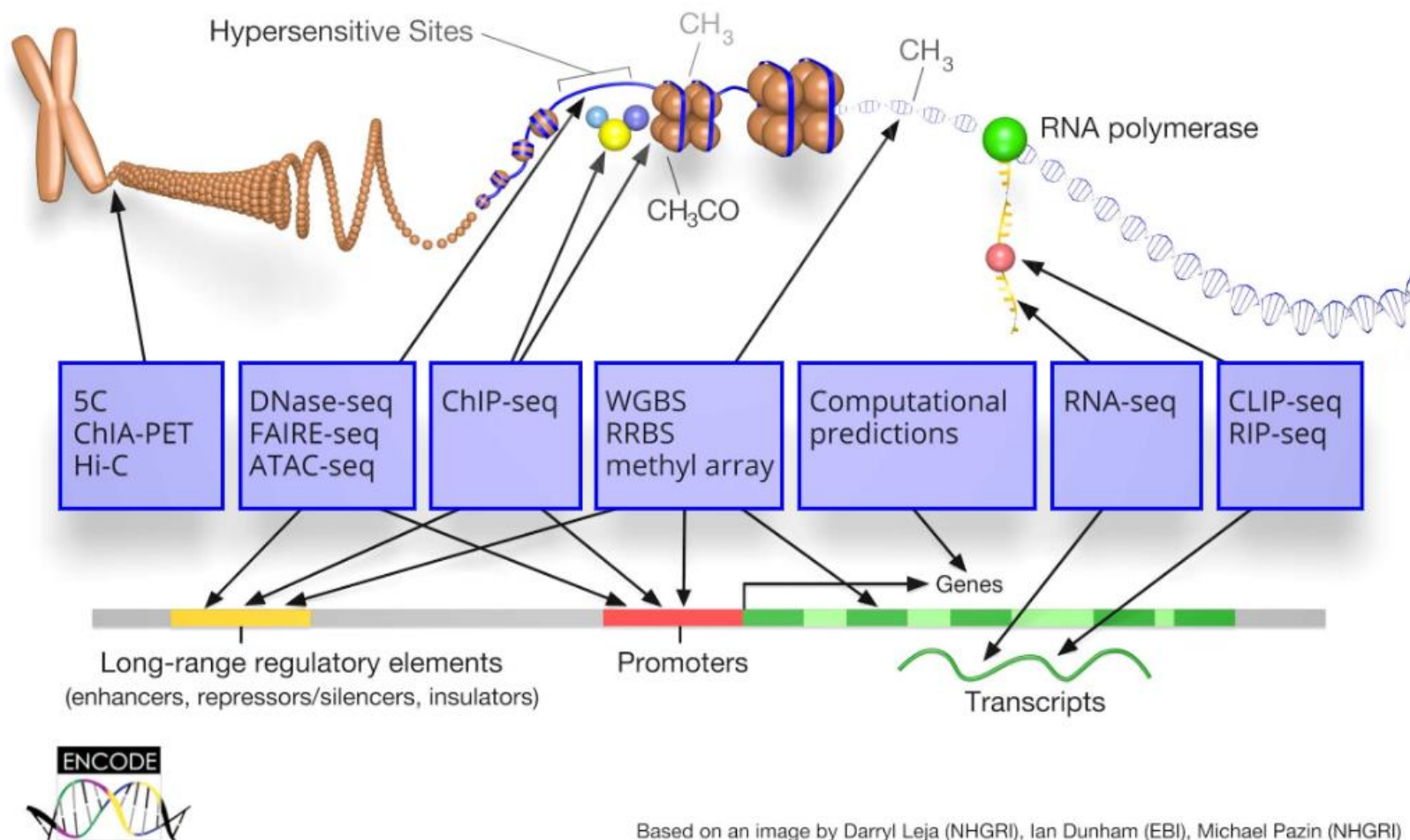




Functional genomics



Functional genomics is the study of how genes and intergenic regions of the genome contribute to different biological processes. (<https://www.ebi.ac.uk/training/online/courses/functional-genomics-i-introduction-and-design/what-is-functional-genomics/>)



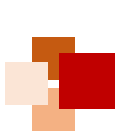
Based on an image by Darryl Leja (NHGRI), Ian Dunham (EBI), Michael Pazin (NHGRI)

<https://www.encodeproject.org/>

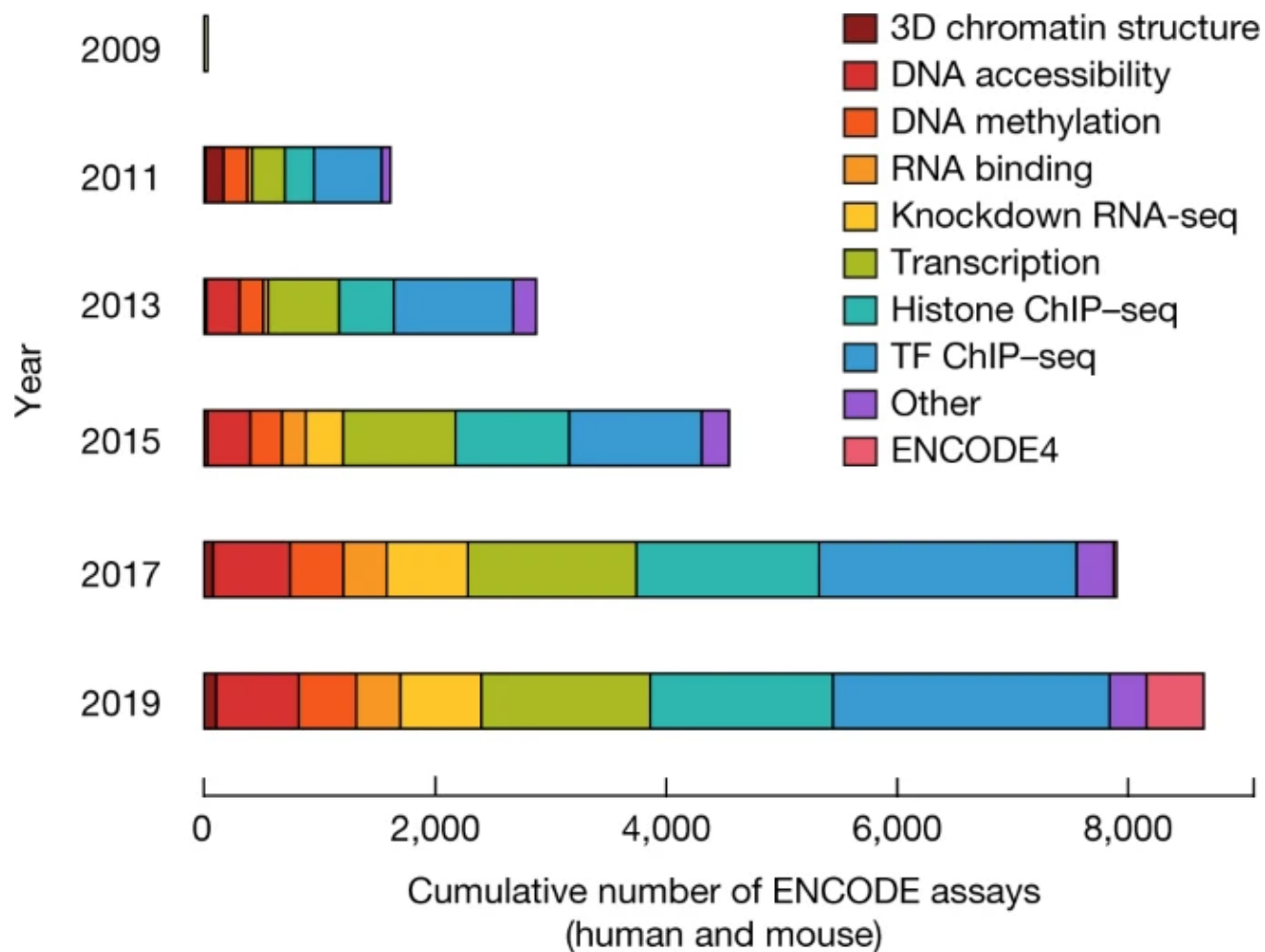
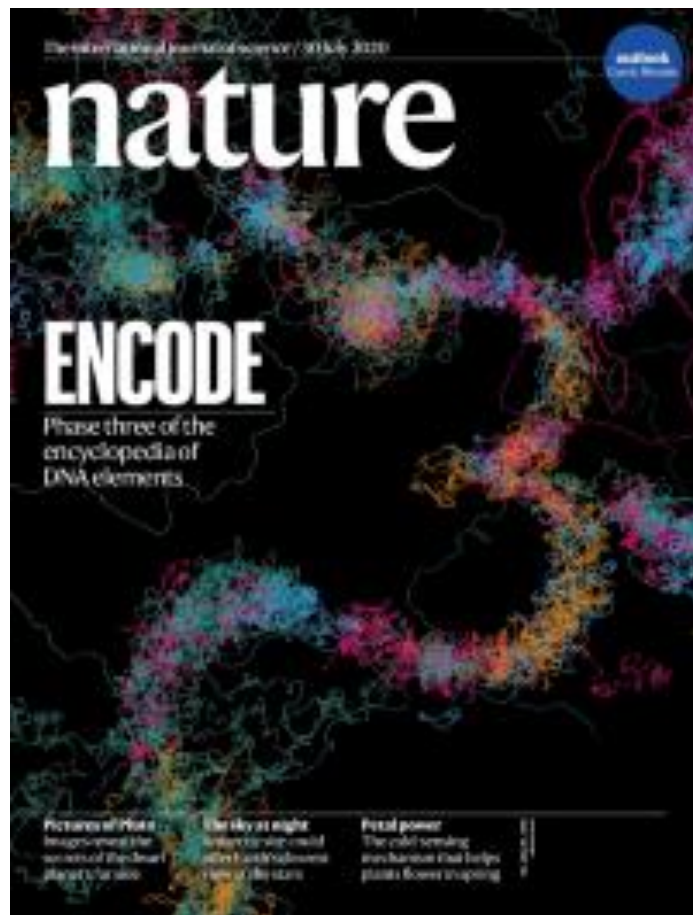
**Bioinformatics tools
evolve with biotech!**

- **Sequence**
- **Quantification**
- **Interaction**
- **Association and causality**





ENCODE on model organisms



Abascal, F., Acosta, R., Addleman, N.J. *et al.* *Nature* 2020; 583:693–698.





CCLE on cancer cell lines

← → ↺

depmap.org/portal/cell_line/ACH-000364?tab=mutation

应用

统一身份认证

学术

学校和所资源

statistics

database

BIOINFORMATICS

NGS experiments

researchers

vendors for device

animals

E-supply

bioinformatics

其他杂

depmap portal

Enter a gene, cell line, lineage or compound

Tools

Downloads

About

Questions? Feedback?

U2OS

Depmap ID: ACH-000364 CCLL Name: U2OS_BONE COSMIC ID: 909776 Sanger ID: 1145 Cellosaurus RRID: CVCL_0042
Aliases: U-2 OS, U2-OS, U-2 OS

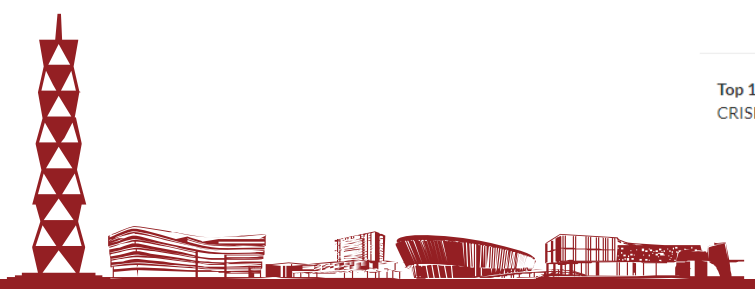
Disease	Lineage	Primary/Metastasis
Bone Cancer	Bone	Primary
Disease Subtype	Lineage Subtype	
Osteosarcoma	Osteosarcoma	
Source	Gender	
ATCC	Female	

U2OS sequenced and profiled in following datasets

Loss-of-function screens	Omics Characterizations	Compound Viability screens
CRISPR (Avana) Public 20Q3	Expression Public 20Q3	Drug sensitivity AUC (Sanger GDSC1)
Combined RNAi (Broad, Novartis, Marcotte)	Copy Number Public 20Q3	Drug sensitivity AUC (Sanger GDSC2)
RNAi (Broad)	Mutation Public 20Q3	Drug sensitivity IC50 (Sanger GDSC1)
RNAi (Novartis)	Fusion 20Q3	Drug sensitivity IC50 (Sanger GDSC2)
	Translocation	Drug sensitivity AUC (CTD^2)
	Protein Array	Drug sensitivity (PRISM Repurposing Primary Screen) 19Q4
	Methylation (1kb upstream TSS)	Drug sensitivity AUC (PRISM Repurposing Secondary Screen) 19Q4
	Copy Number (Absolute)	Drug sensitivity replicate-level dose (CTD^2)
	Proteomics	Drug sensitivity replicate-level dose (Sanger GDSC1)
		Drug sensitivity replicate-level dose (Sanger GDSC2)
		Drug sensitivity dose-level (PRISM Repurposing Secondary Screen) 19Q4
		Drug sensitivity replicate-level dose (PRISM Repurposing Secondary Screen) 19Q4

Top 10 preferentially essential genes	
CRISPR (Avana) Public 20Q3	Combined RNAi (Broad, Novartis, Marcotte)

<https://depmap.org/portal/>

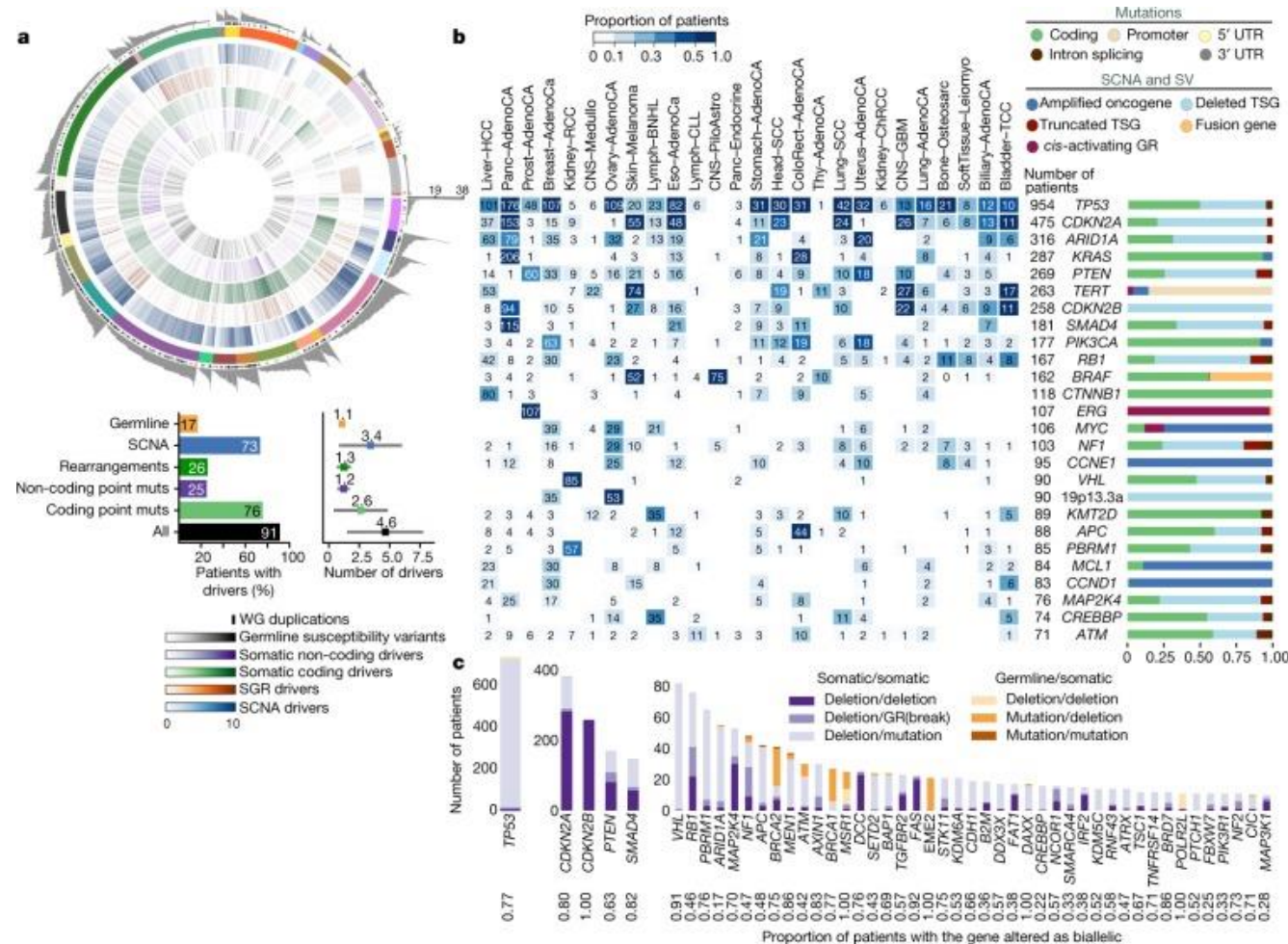
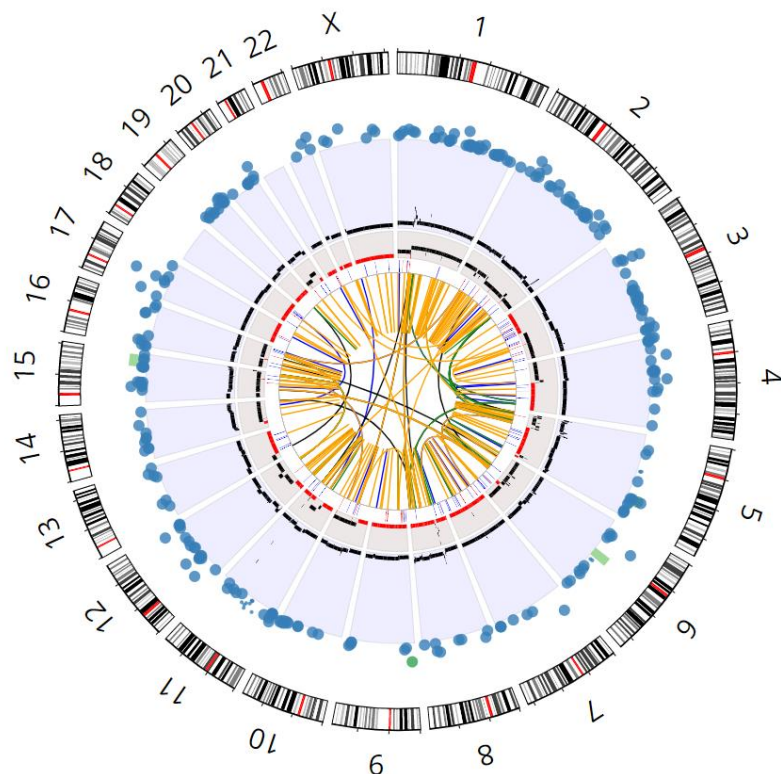




Cancer genomics project



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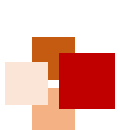


<http://compbio.med.harvard.edu/chromothripsis/>

<https://cancer.sanger.ac.uk/cosmic>

The ICGC/TCGA Pan-Cancer Analysis of Whole Genomes Consortium.
[Nature](#). 2020; 578(7793): 82–93. Pan-cancer analysis of whole
genomes

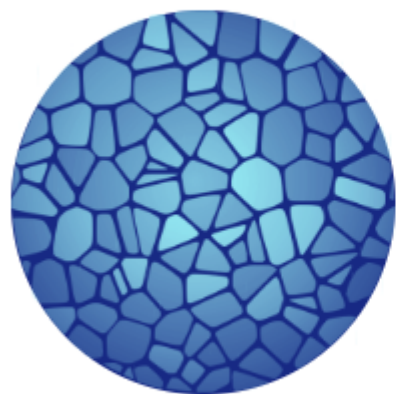
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Human cell atlas



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**HUMAN
CELL
ATLAS**

A map of human cells

Mission:

To create comprehensive reference maps of all human cells—the fundamental units of life—as a basis for both understanding human health and diagnosing, monitoring, and treating disease

<https://www.humancellatlas.org/>



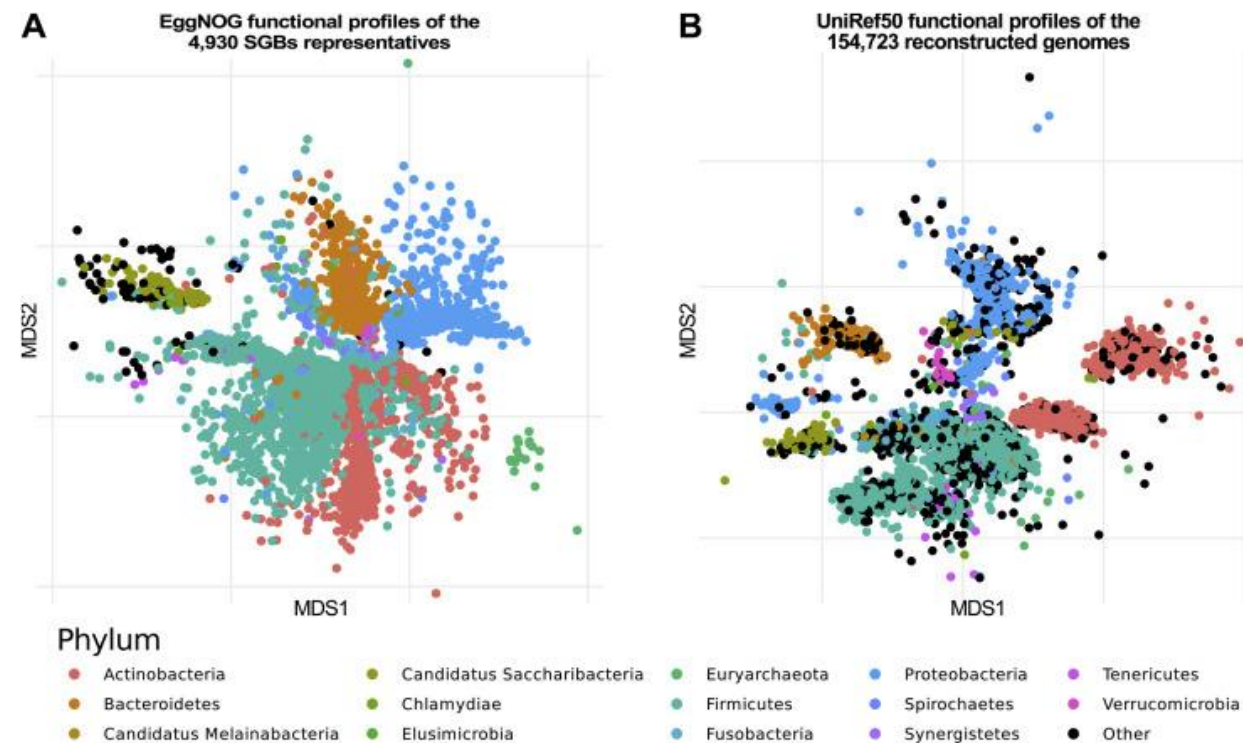
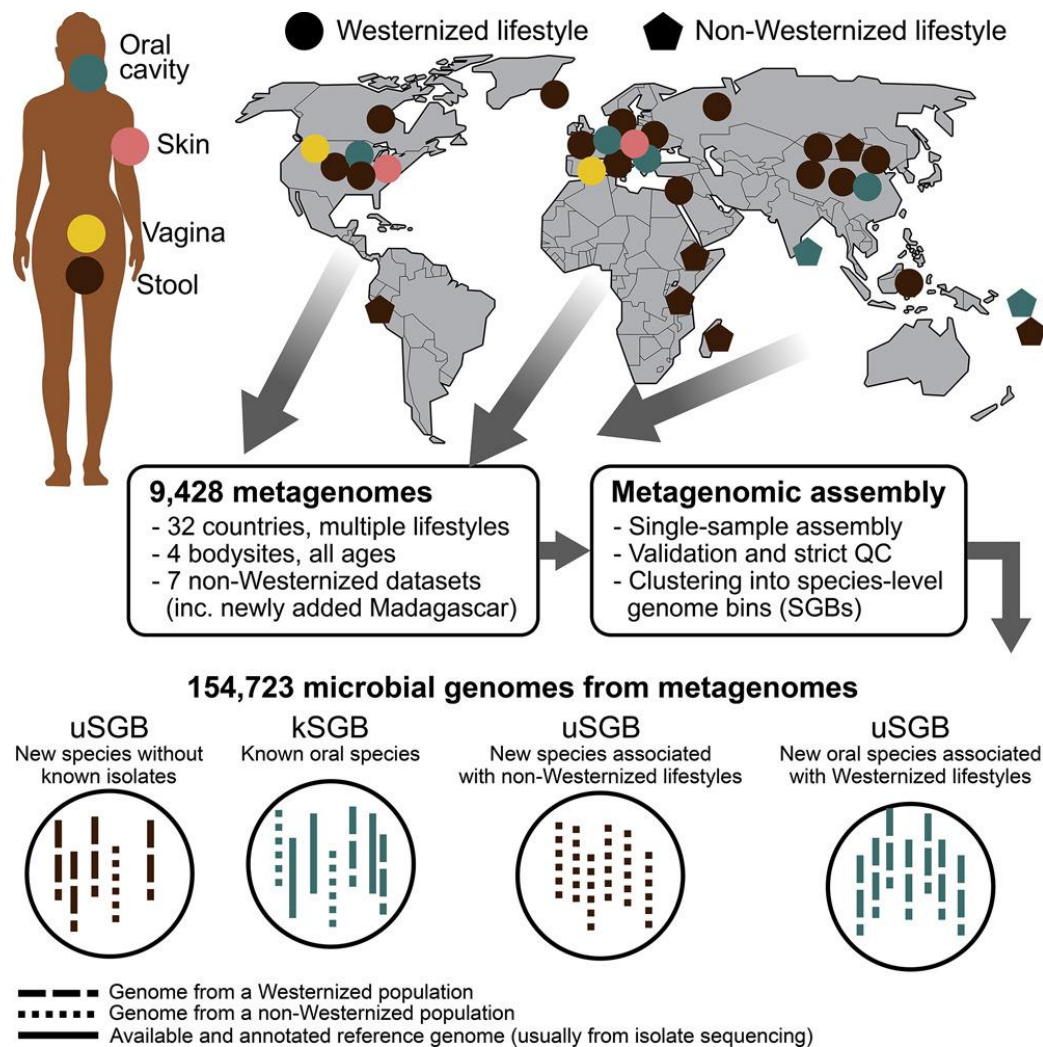
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Metagenomics



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Fan Wu et al. Nature. 2020; 579(7798): 265–269

<https://idseq.net/>

<http://merenlab.org/2020/07/27/history-of-metagenomics/>

Cell (2019) 176(3):649-662



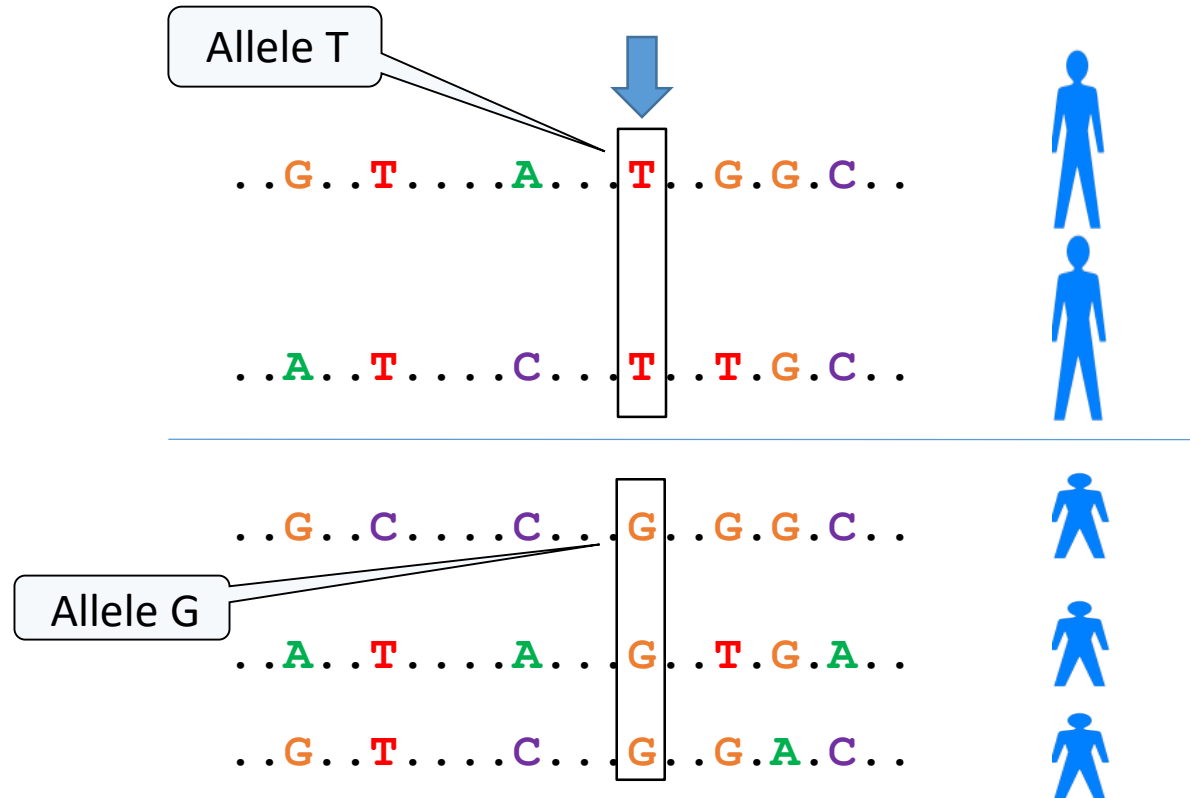
Genome-Wide Association Studies (GWAS)



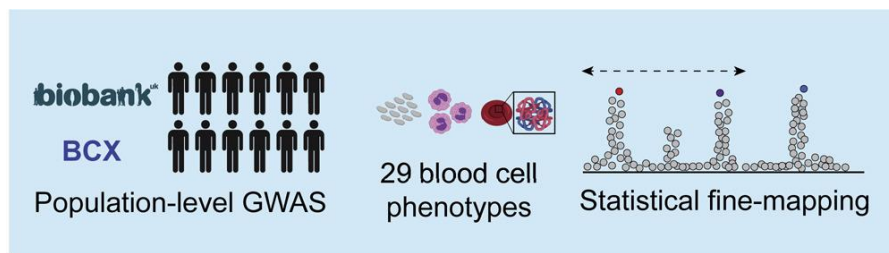
Quantitative Trait Loci (QTL) Mapping



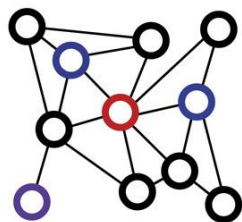
- Association between
 - Single Nucleotide Polymorphism (SNPs)
 - Phenotypic variation (e.g. height)



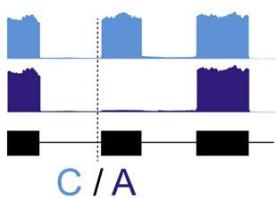
Human GWAS study



Omnigenic/Infinitesimal model evaluation



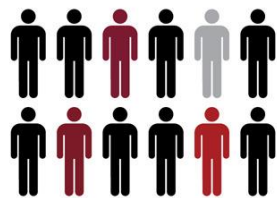
Functional variant interpretation



Cell state chromatin mapping



Polygenic modification of blood disease risk



- Largest genome-wide association study (563,085) of blood cell traits to date
- Empiric assessments of omnigenic and infinitesimal models of polygenic variation
- Functional insights into how genetic variants impact human hematopoiesis
- Assessment of the effect of polygenic trait scores upon blood diseases

Dragana Vuckovic et al. Cell (2020), 182(5):1214-1231

- **Genome-wide association studies (GWAS)** uses SNP markers to identify disease loci
 - In **family-based** designs, markers are measured in a **proband** (i.e. the affected individual) and unaffected individuals in a family to identify the differences in the frequency of variants
 - In **population-based** designs, a large number of **unrelated** cases and controls are studied (typically hundreds or thousands in each group). Larger sample sizes increase statistical power
- It is difficult to identify the genetic causes of common human diseases that involve **multiple** genes, each of which may make only a **small** contribution to the disease risk

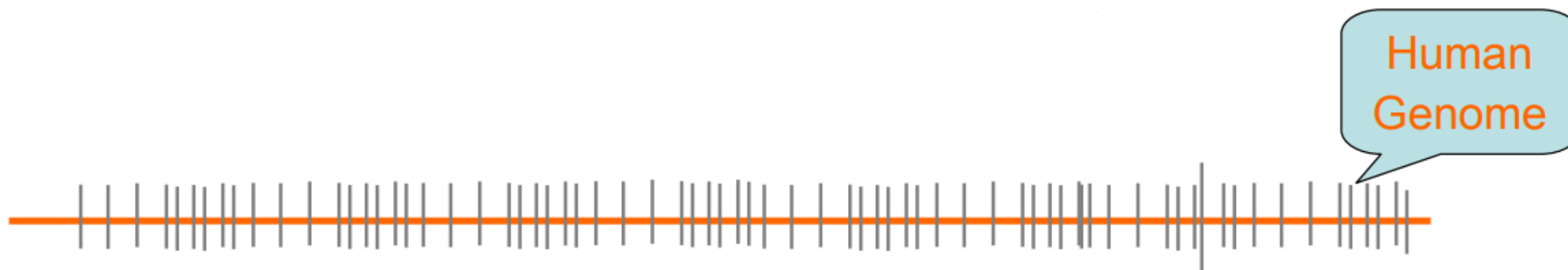




Definition of GWAS



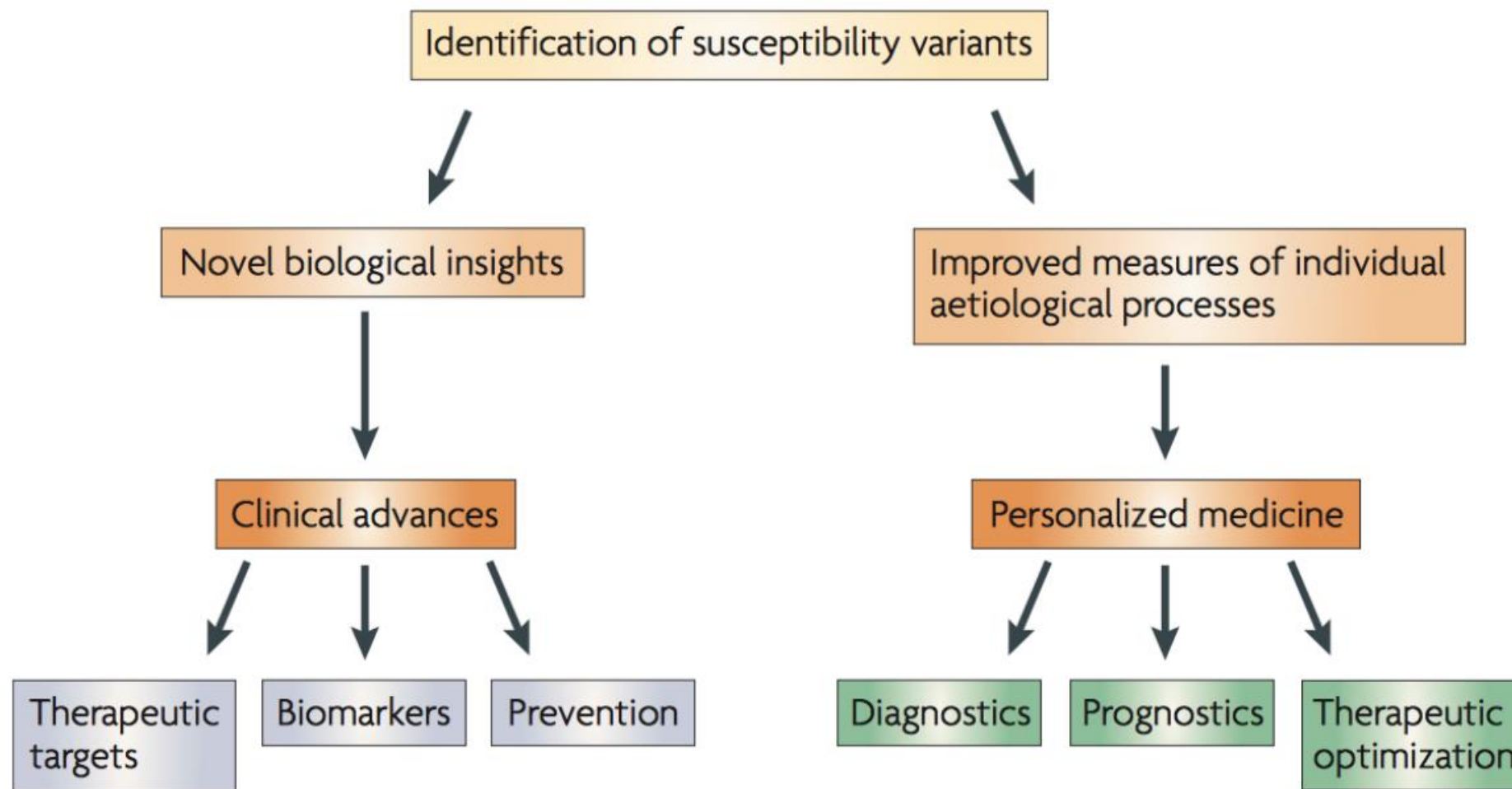
- GWAS is the association analysis performed with a panel of **polymorphic markers** (e.g. SNPs) adequately spaced to capture most of the **linkage disequilibrium (LD)** information in the entire genome in the population under study. Usually, a GWAS involves 500k – 4m SNPs



Test each SNP for association with disease phenotype



Clinical translation of findings from GWAS



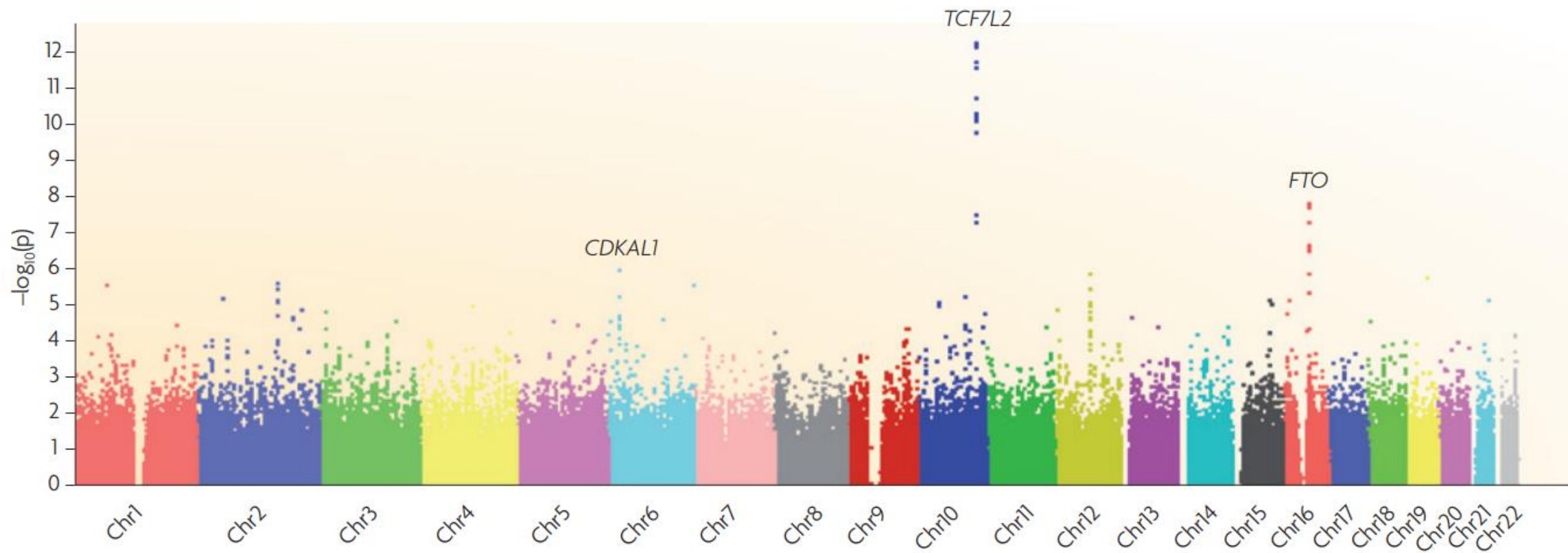
McCarthy MI et al. Genome-wide association studies for complex traits: consensus, uncertainty and challenges. Nature Review Genetics, 2008 May; 9(5):356-69.



Manhattan plots



- Manhattan plots display GWAS findings with respect to the genomic locations of SNPs, highlighting signals of particular interest
- The following is a Manhattan plot for type 2 diabetes





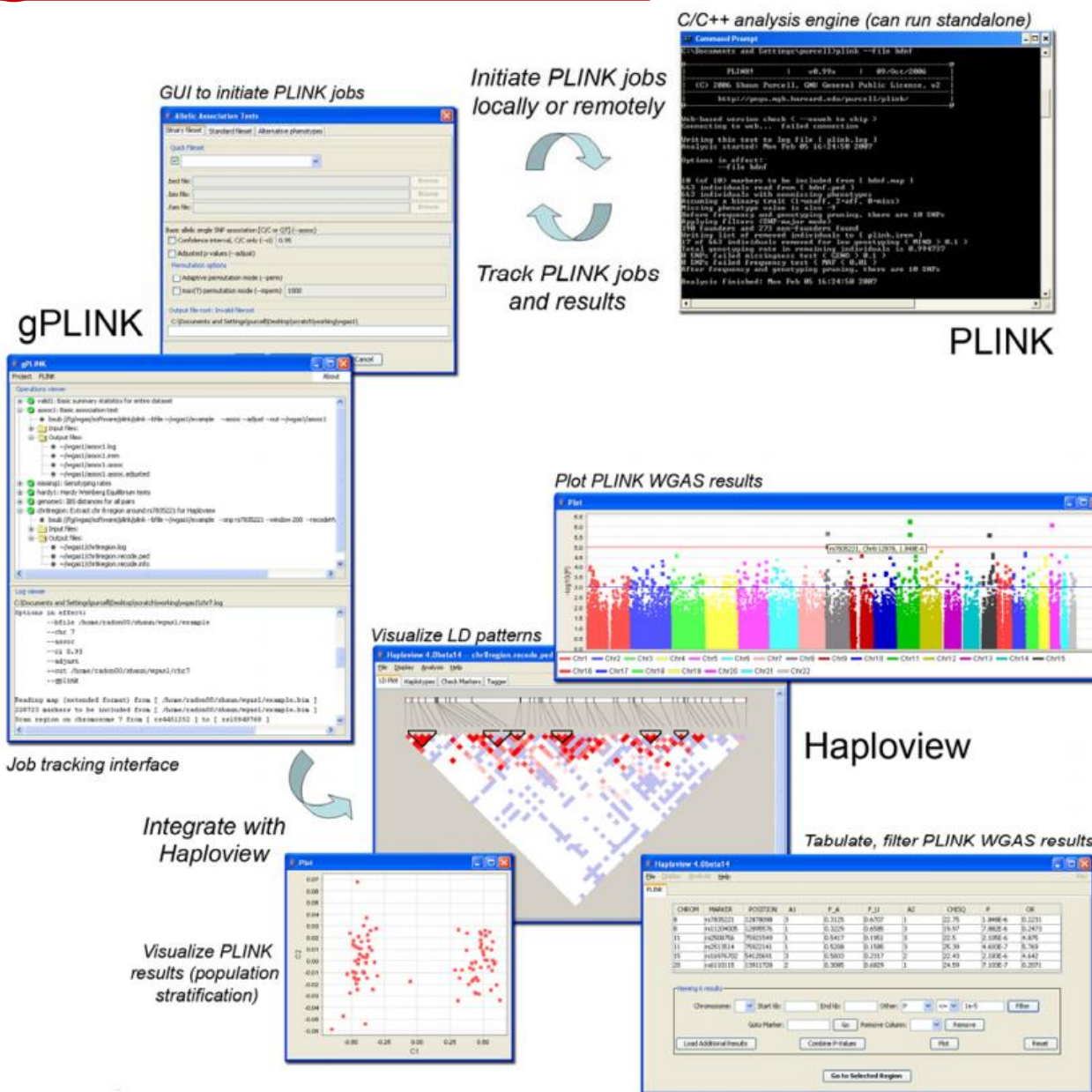
Software case study: PLINK



- PLINK is a free, open-source software toolset for whole-genome association analysis, designed to perform a range of basic, large-scale analyses in a computationally efficient manner
- Available at:
 - <http://zzz.bwh.harvard.edu/plink/> (old version)
 - <https://www.cog-genomics.org/plink/2.0/> (version 2.0)
- PLINK has numerous useful features for managing and analyzing genetic data
 - Data management
 - Summary statistics
 - Population stratification
 - Association analysis
 - Linkage disequilibrium and haplotype analysis
 - Shared segment analysis
 - Copy number analysis



Running PLINK with GUI



From Purcell et al. 2007

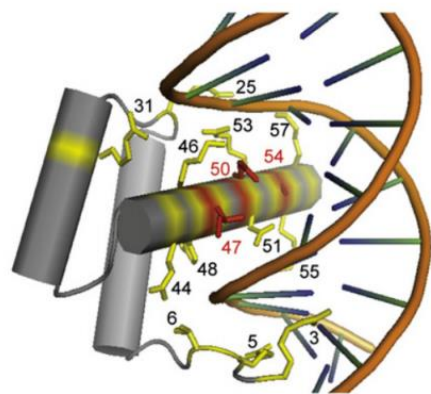
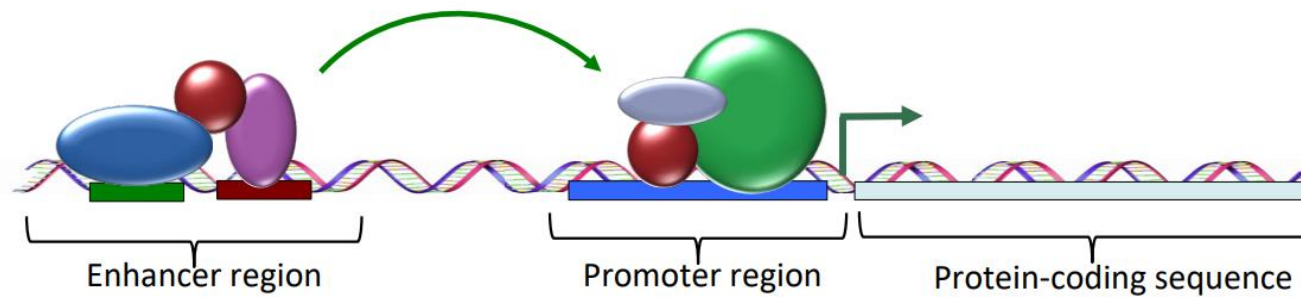


Case Study of Deep Learning in Genomics: DeepBind

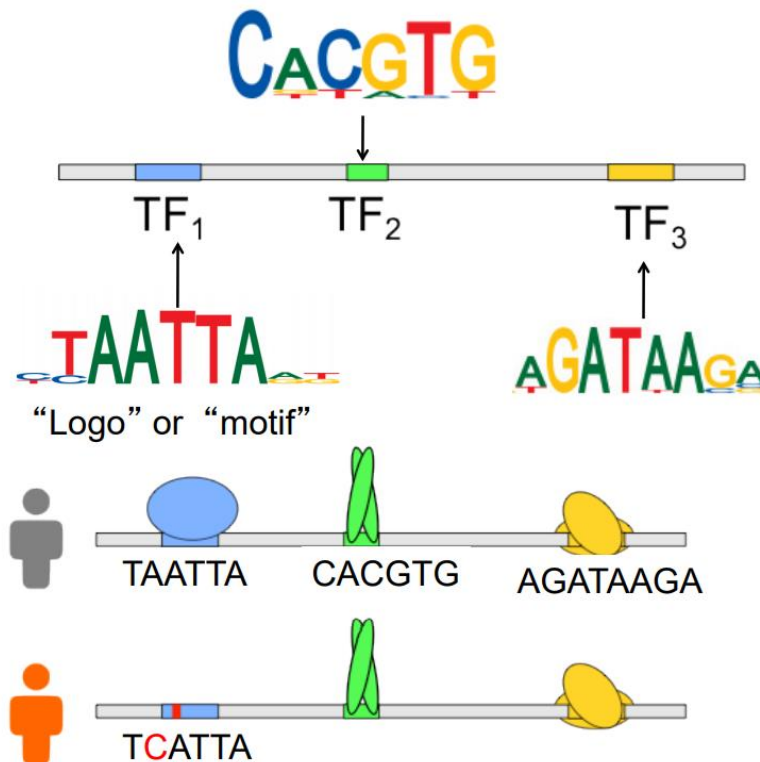
PPT slides adapted from MIT course “Deep Learning in the Life Sciences”, Lecture 6 “Regulatory genomics”, by Prof. Manolis Kellis



Transcription factors (TFs) bind to specific DNA sequences to control gene expression



DNA-binding domain of *Engrailed*

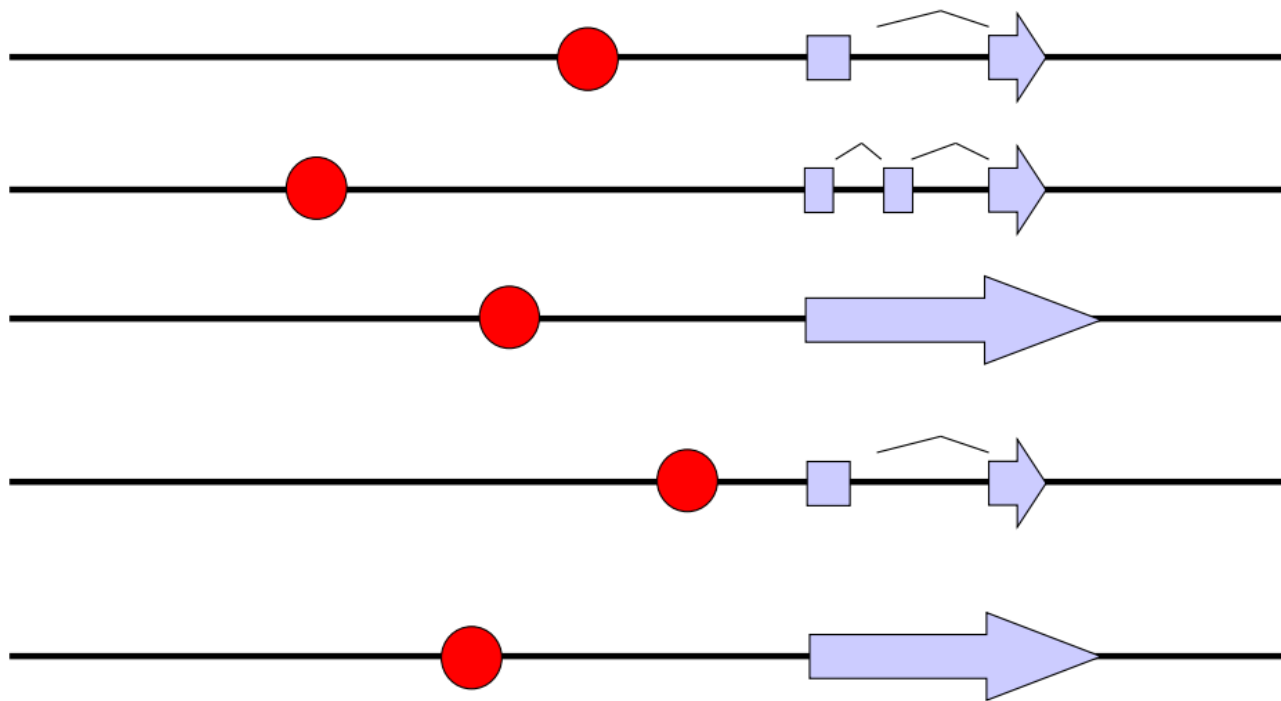




Enrichment-based motif discovery methods



Given a set of **co-regulated/functionally related** genes,
find common motifs in their promoter regions



- Align the promoters to each other using local alignment
- Use expert knowledge for what motifs should look like
- Find “median” string by enumeration (motif/sample driven)
- Start with conserved blocks in the upstream regions



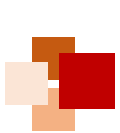


Starting positions $\Leftrightarrow$ motif matrix

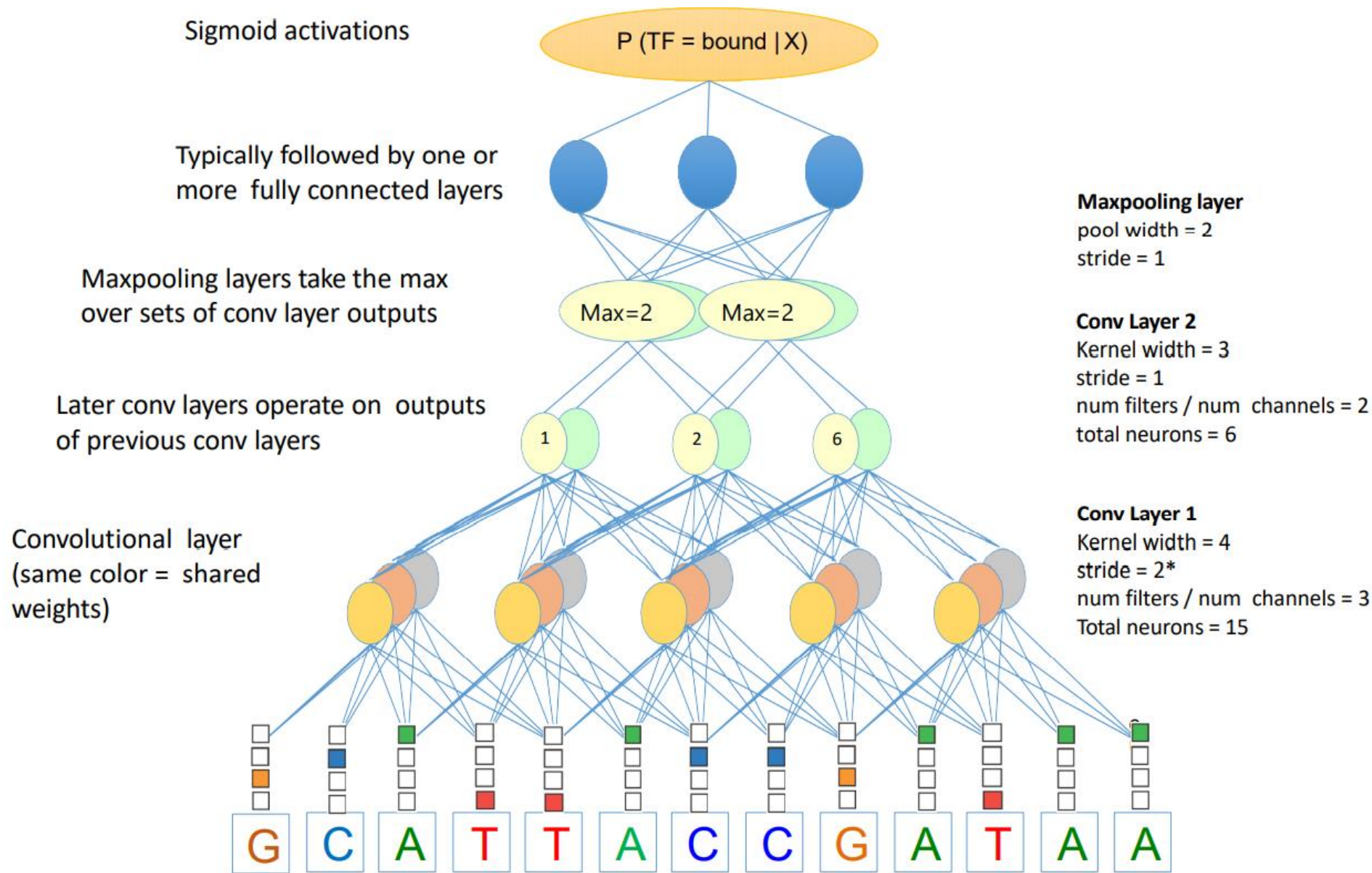


- Given aligned sequences, it is easy to compute a profile matrix
- Given a profile matrix, it is easy to find starting position probability





Deep convolutional neural network



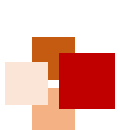


Representing DNA sequence as 2D matrix



Matrix representation of
DNA sequence
(darker = stronger)

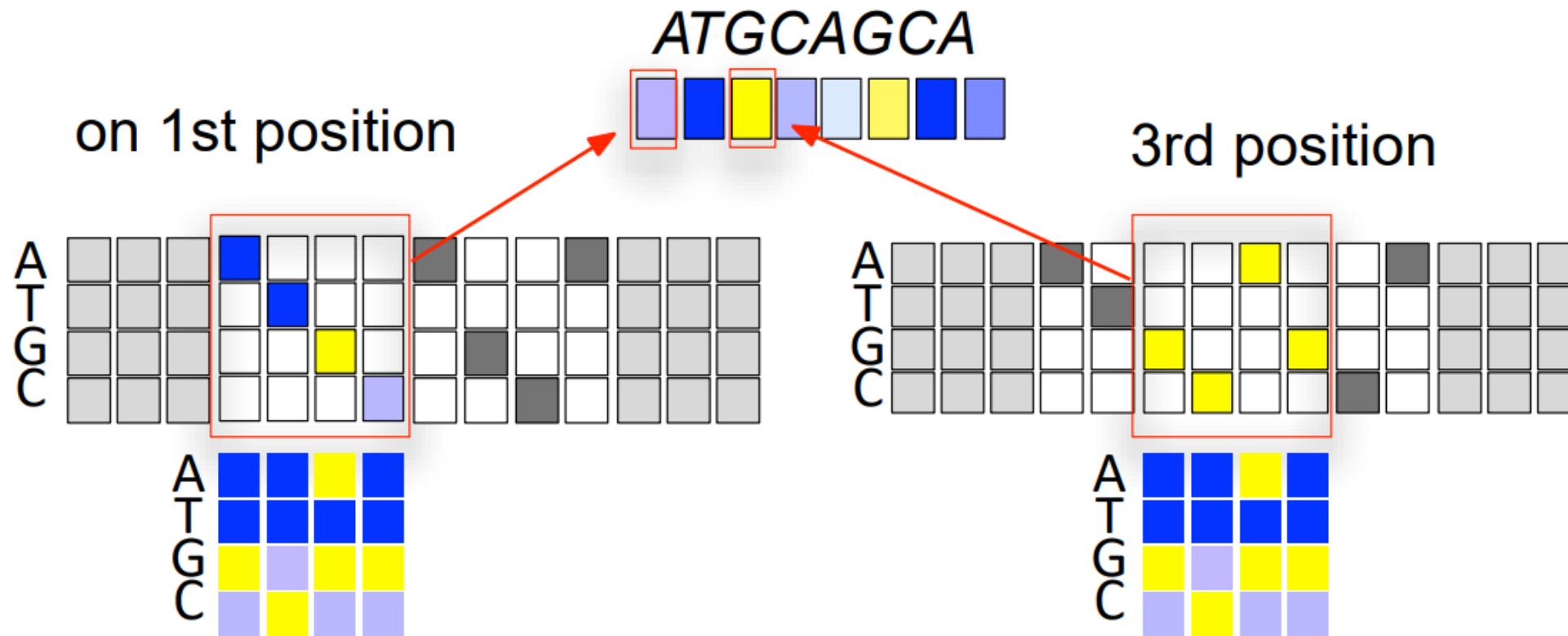




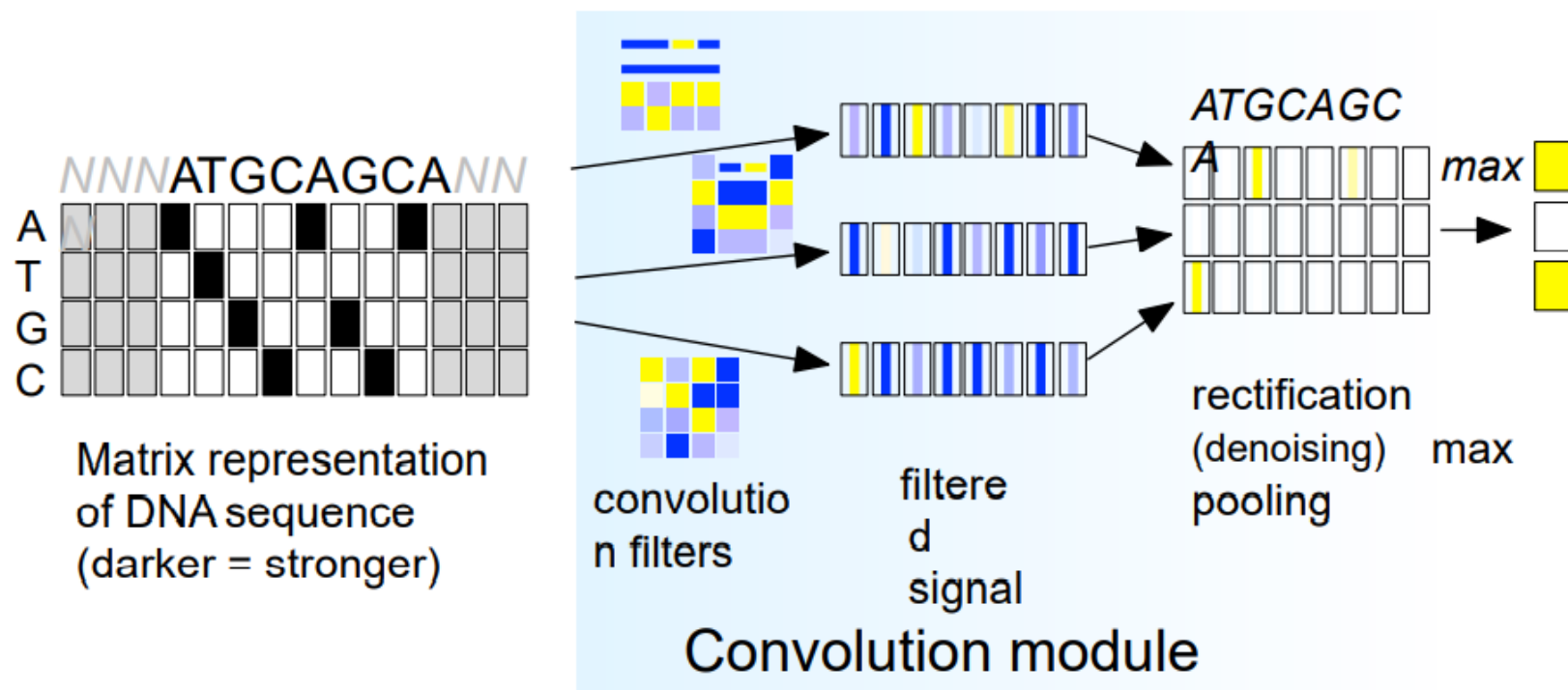
Convolution – extracting invariant feature



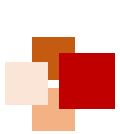
Applying 4 bp sequence filter along the DNA matrix:



Convolution – extracting invariant feature (continued)



Rectification = ignore signals below some threshold.
Pooling = summary of each channel by max or average.

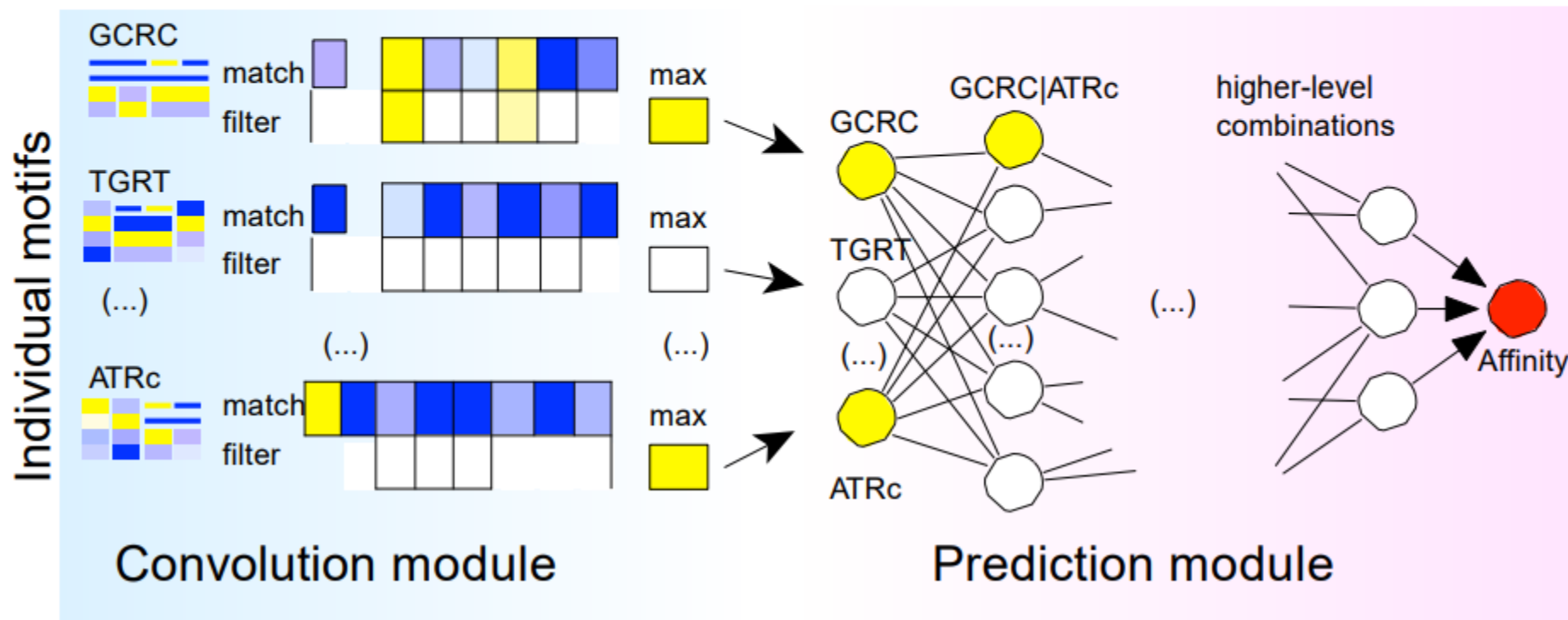


Prediction using extracted feature map



ChIP-seq, PBMs, SELEX Experiments DNA sequence

	A	T	G	C	A	G	C	A	N	N	N
A	Black				Black			Black	Gray	Gray	Gray
T		Black							Gray	Gray	Gray
G			Black			Black			Gray	Gray	Gray
C				Black			Black		Gray	Gray	Gray



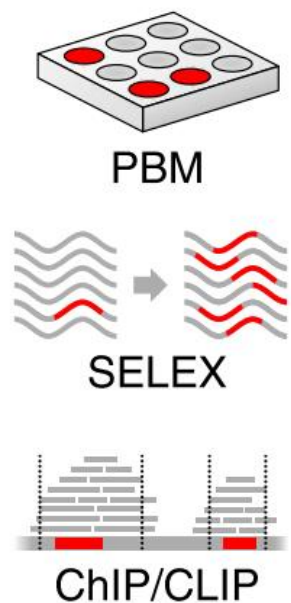


DeepBind – overview of pipeline

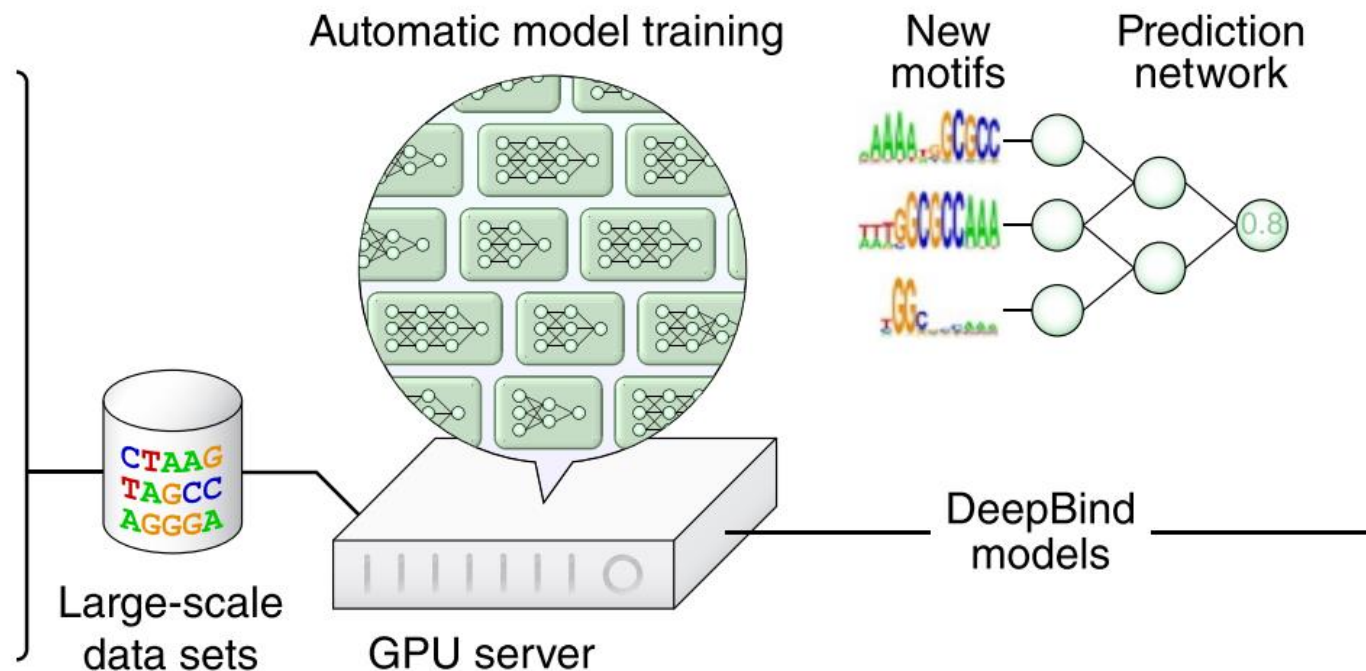


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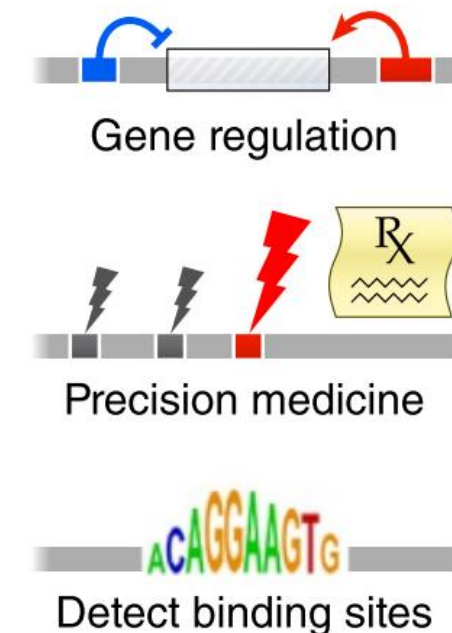
1. High-throughput experiments



2. Massively parallel deep learning

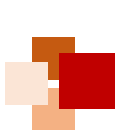


3. Community needs



Alipanahi et al. Nature Biotechnology 2015

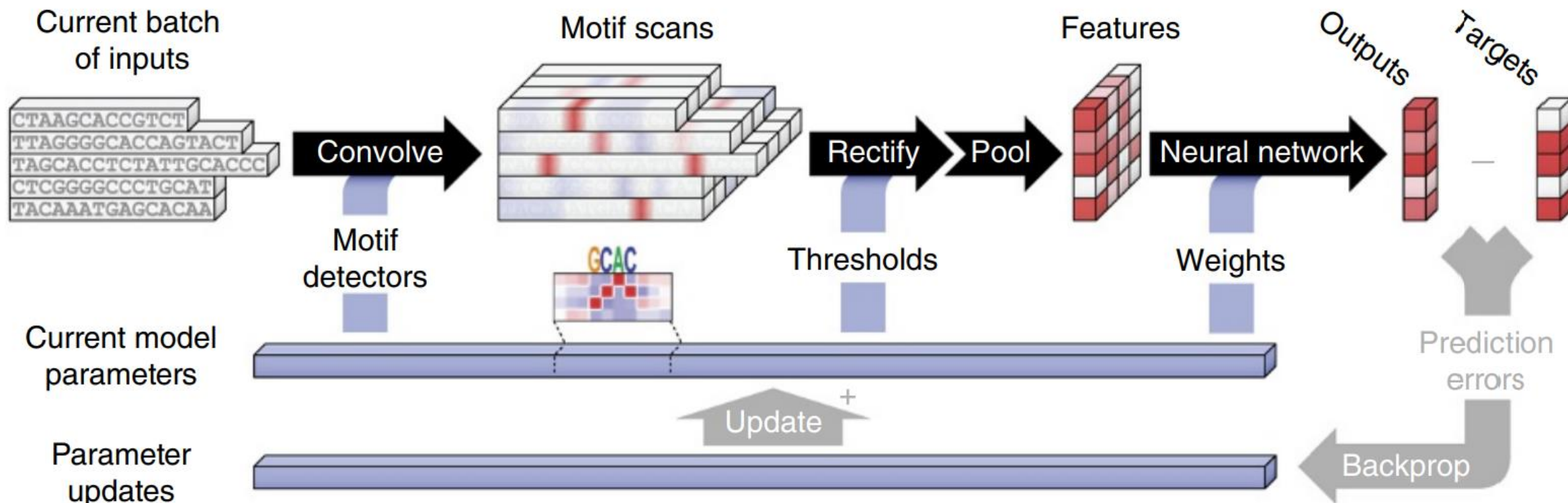




DeepBind – details of inner workings



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Alipanahi et al. Nature Biotechnology 2015





Recap



- In this lecture we have learned:
 - Basic concepts and brief history of genomics
 - Technologies for DNA sequencing, especially the Next-Generation Sequencing (NGS)
 - Basic steps in a pipeline of GWAS
 - How to use PLINK software for GWAS
 - Key ideas of using CNN for DNA sequencing modeling
 - How DeepBind works to predict the binding specificities between proteins and DNA/RNA sequences





References



- Jonathan Pevsner (2015). Bioinformatics and Functional Genomics. (3rd Edition):
 - Chapter 9 "Analysis of Next-Generation Sequencing Data"
 - Chapter 20 "Human Genome"
 - Chapter 21 "Human Disease"
- Purcell, S. et al. PLINK: A tool set for whole-genome association and population-based linkage analyses. American Journal of Human Genetics, Vol. 81, pp. 559-575 (2007).
- Shendure, J. et al. DNA sequencing at 40 past, present and future. Nature 550(7676):345-353 (2017).
- Sara Goodwin et al. (2016) Coming of age: Ten years of next-generation sequencing technologies, Nature Reviews Genetics, 17(6), 333-351.
- Sapoval, N., Aghazadeh, A., Nute, M.G. et al. Current progress and open challenges for applying deep learning across the biosciences. Nature Communications **13**, 1728 (2022).
- Zou, J. et al. A primer on deep learning in genomics. Nature Genetics **51**, 12-18 (2019).
- Alipanahi, B. et al. Predicting the sequence specificities of DNA- and RNA-binding proteins by deep learning. Nature Biotechnology, **33**, 831-838 (2015). (**DeepBind**)





Appendix: Statistical Modeling in GWAS



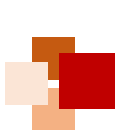


Case-control study design



- Cases:
 - Ascertained for the phenotype of interest
 - Presumed to have a high prevalence of susceptibility alleles
- Controls:
 - Not ascertained for the phenotype
 - Presumed to have lower prevalence of such susceptibility alleles
- Misclassification bias: e.g. failure to identify **latent diagnoses** in the control group. It leads to loss of power, especially for common traits (e.g. high blood pressure)





Analysis outline



- Data cleaning / QC (quality control)
- Association test between genotype and phenotype:
 - Analyze each SNP separately
 - Use logistic or linear regression with additive models, including important covariates
 - Calculate a p-value for each SNP
- Control for population substructures, e.g. add principal components as covariates
- Visualization of results
 - Q-Q plots
 - Manhattan plots
- Control for multiple testing issue, e.g. Bonferroni correct, false discovery rate (FDR), Q-value





Odds ratios



- A measure of effect size, or strength of association
- Odds = $P / (1 - P)$, where P stands for probability of an event
- **Odds ratio** is the ratio of odds for an event (having a disease) under two comparing conditions (e.g. different genotypes at a SNP)
- Example (likelihood of a disease):
 - $P(D \mid \text{genotype "AT"}) = 0.8$
 - $P(D \mid \text{genotype "TT"}) = 0.2$
 - The odds ratio for getting the disease with genotype AT compared to TT is
$$OR = (0.8/0.2) / (0.2/0.8) = 16$$
 - The OR for AT individuals relative to an average population risk of 25% is:
$$OR = (0.8/0.2) / (0.25 / 0.75) = 12$$



Odds ratios in GWAS



- GWAS would often report ORs relative to the low-risk allele or lowest-risk genotype
- To turn this into a meaningful risk estimate, the **prevalence of the disease** and the **genotype frequencies** must be taken into account
- $P(D) = \text{prevalence}$
$$= P(D|AA)P(AA) + P(D|Aa)P(Aa) + P(D|aa)P(aa)$$
- For each SNP, an odds ratio and associated p-value are calculated (e.g. using the chi-square test)
- Significance of the association is measured by: $-\log_{10}(P\text{-value})$





Multiple hypothesis testing issue



- A conventional $\alpha = 0.05$ threshold for p-values assumes a 5% chance of false-positive finding due to chance
 - When performing only one test, this is reasonable
 - However, when performing many (e.g. 1,000,000) tests, which is common in GWAS, this will lead to many false positives
- Example: Consider a case with 20 hypotheses to test, and the threshold of significant p-value is $\alpha = 0.05$. The probability of observing at least one significant result is:

P(at least one significant result)

$$= 1 - P(\text{no significant results}) = 1 - (1 - 0.05)^{20} \approx 0.64$$

- So, with 20 tests being considered, there is a 64% chance of observing at least one “significant” result, even if none of the tests is actually significant
- Correction methods:
 - **Bonferroni correction**: sets the significance cut-off at α/n , where n is the number of tests
 - **False discovery rate (FDR)**: expected proportion of false positives among all results claimed as significant
 - **Q-value**: The minimum FDR at which the test may be called significant, serving as a p-value that has been adjusted for the FDR

